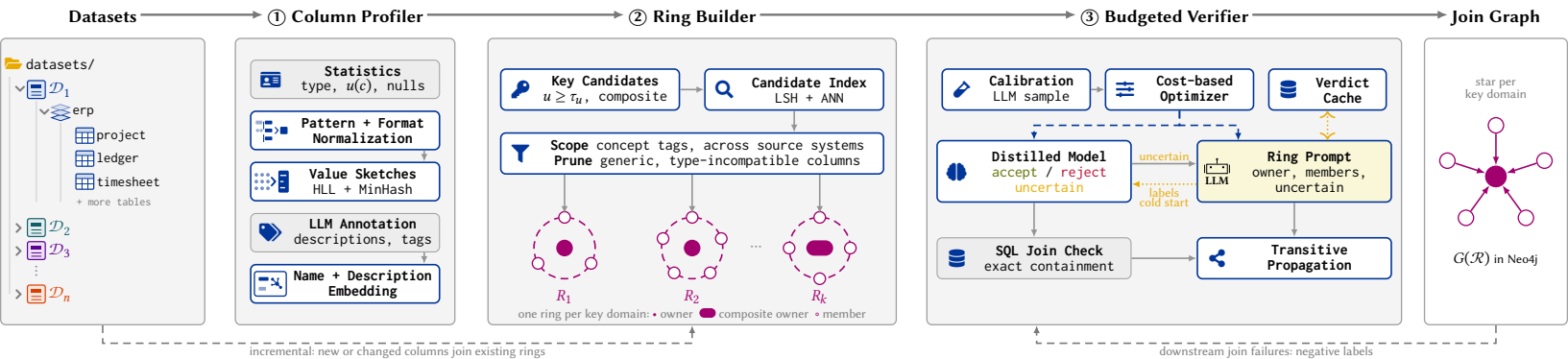

Datasets
① Column Profiler
② Ring Builder
③ Budgeted Verifier
Join Graph
datasets/
$\mathcal{D}_1$
erp
project
ledger
timesheet
+ more tables
$\mathcal{D}_2$
$\mathcal{D}_3$
$\mathcal{D}_n$
Statistics
type, $u(c)$, nulls
Pattern + Format Normalization
Value Sketches
HLL + MinHash
LLM Annotation
descriptions, tags
Name + Description Embedding
Key Candidates
$u \geq \tau_u$, composite
Candidate Index
LSH + ANN
Scope concept tags, across source systems
Prune generic, type-incompatible columns
R_1
R_2
R_k
one ring per key domain: • owner composite owner ◦ member
Calibration
LLM sample
Cost-based Optimizer
Verdict Cache
Distilled Model
accept / reject
uncertain
uncertain
labels
cold start
LLM
Ring Prompt
owner, members, uncertain
SQL Join Check
exact containment
Transitive Propagation
star per key domain
$G(\mathcal{R})$ in Neo4j
incremental: new or changed columns join existing rings
downstream join failures: negative labels
new KeyRing component
existing pipeline component